

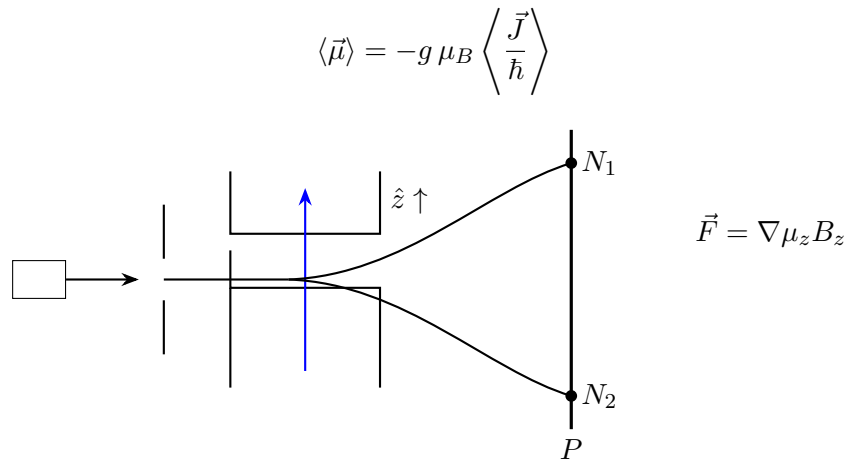
Quantum Optics — Lecture 2

Two-level atomic systems: Stern–Gerlach and spin $\frac{1}{2}$

Transcribed from handwritten notes (Note Sep 15, 2026, pp.3–7)

Two level atomic systems

Stern–Gerlach + Spin $\frac{1}{2}$



Interaction energy of the magnetic moment with the magnetic field is just $-\vec{\mu} \cdot \vec{B}$.

$$F_z = \frac{\partial}{\partial z} (\vec{\mu} \cdot \vec{B}) \simeq \mu_z \frac{\partial B_z}{\partial z}$$

- If $\mu_z > 0$ ($S_z < 0$) the atom experiences an upward force.
- If $\mu_z < 0$ ($S_z > 0$) the atom experiences a downward force.
- The atoms in the furnace are randomly oriented. So classically we would expect values between $|\mu|$ and $-|\mu|$; this would lead to a spread in the values, unlike the only two spots being observed.

$\Rightarrow$ Later this was realized to be spin.

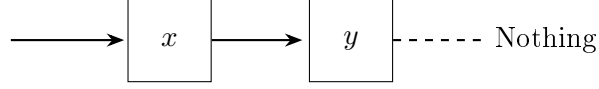
$\Rightarrow$ We also see the weird measurement effects with multiple Stern–Gerlach experiments.

$$\vec{E} = E_0 \hat{x} \cos(kz - \omega t)$$

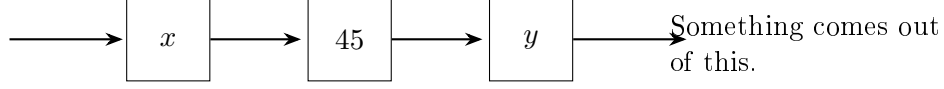
Likewise we may consider y -polarized light, also propagating in the z -direction

$$\vec{E} = E_0 \hat{y} \cos(kz - \omega t)$$

Then we have that



but



Constructing spin operators from sequential SG experiments

We know we can write

$$S_z = \frac{\hbar}{2} \left[|+\rangle \langle +| - |-\rangle \langle -| \right]$$

Classically a current loop has the magnetic moment

$$\begin{aligned} \mu &= \frac{IA}{c} = \frac{qA}{Tc} = \frac{qvA}{2\pi r c} = \frac{qv \pi r^2}{2\pi r c} \\ &= \frac{qv}{2c} r = \frac{q}{2mc} mvr = \frac{q}{2mc} L \end{aligned}$$

so

$$\mu = \frac{q}{2mc} L.$$

For spin $\mu = g \frac{q}{2mc} S$; $g = 1 \Rightarrow$ classical orbital value. Dirac says $g = 2$:

$$\mu = \frac{q}{mc} S.$$

$$|S_z| = \frac{m_e c}{e} |\mu_z|, \quad \mu_z = \frac{e\hbar}{2m_e c} (\text{exp.}) \quad \Rightarrow \quad S_z = \frac{\hbar}{2}.$$

We have that

$$S_z = \frac{\hbar}{2} \left[|+\rangle \langle +| - |-\rangle \langle -| \right]$$

Obs: the S_x beam splits 50/50 in $\text{SG}\hat{z}$, so we require that

$$|\langle +|S_x; +\rangle|^2 = |\langle -|S_x; -\rangle|^2 = \frac{1}{2}$$

but it also must be the case that

$$|S_x; +\rangle = c_1 |+\rangle + c_2 |-\rangle,$$

thus combining the two and taking into account an unphysical phase we have that

$$|S_x; +\rangle = \frac{1}{\sqrt{2}} |+\rangle + \frac{e^{i\delta_1}}{\sqrt{2}} |-\rangle.$$

Then since $|S_x; -\rangle$ is necessarily orthogonal to $|S_x; +\rangle$ we have that

$$|S_x; -\rangle = \frac{1}{\sqrt{2}} |+\rangle - \frac{e^{i\delta_1}}{\sqrt{2}} |-\rangle.$$

Obs: S_y also splits in SGx, so

$$\begin{aligned}\langle S_y; + | S_x; + \rangle &= \frac{1}{2} (1 + e^{i(\delta_1 - \delta_2)}) \\ |\langle S_y; + | S_x; - \rangle|^2 &= \frac{1}{4} |1 + e^{i(\delta_1 - \delta_2)}|^2 = \frac{1}{2} (1 + \cos(\delta_1 - \delta_2)) = \frac{1}{2} \\ &\iff \cos(\delta_1 - \delta_2) = 0 \iff \delta_2 - \delta_1 = \pm \frac{\pi}{2}\end{aligned}$$

Let $\delta = \frac{\pi}{2}$, so

$$|S_x; \pm\rangle = \frac{1}{\sqrt{2}}(|+\rangle \pm |-\rangle) \quad \text{and} \quad |S_y; \pm\rangle = \frac{1}{\sqrt{2}}(|+\rangle \pm i|-\rangle).$$

So

$$\begin{aligned}S_x &= \frac{\hbar}{2} (|S_x; +\rangle \langle S_x; +| - |S_x; -\rangle \langle S_x; -|) \\ &= \frac{\hbar}{2} (|+\rangle \langle -| + |-\rangle \langle +|) \\ S_y &= \frac{\hbar}{2} (-i|+\rangle \langle -| + i|-\rangle \langle +|).\end{aligned}$$